

RESEARCH STATEMENT

My research spans two areas: Trustworthy AI and NeuroAI. Within **Trustworthy AI**, I focus on *AI Interpretability* (ICML2026 Oral, ICML2026 Mech. Interp. Workshop), probing memory traces in LLMs, *AI Steerability* (ICLR2026, ICLR2026), editing and unlearning model knowledge, and *AI Alignment* (AAAI2026, ACL2026 Findings), studying human-LLM decision alignment. Within **NeuroAI**, I pursue *Neuroscience for AI* (NeurIPS2023, ICLR2025), translating brain mechanisms into model architectures, and *AI for Neuroscience* (IJCV2024, Exp.Mol.Med.2026), applying ML to behavioral and neural data.

HIGHLIGHTED PUBLICATIONS

Trustworthy AI

ICML 2026 (*Oral*)

First

AI Engram: In Search of Memory Traces in Artificial Intelligence

Kwon et al.

ICML 2026 (*Mech. Interp. Workshop*)

First

Moir: Let the Model Direct Its Own Story for Robust Cross-Domain Knowledge Editing

Kwon et al.

ICLR 2026

Co-author

Bilinear Relational Structure Fixes Reversal Curse and Enables Consistent Model Editing

Kim, Kwon et al.

ICLR 2026

Co-author

Erase or Hide? Suppressing Spurious Unlearning Neurons for Robust Unlearning

Yang, Kwon et al.

AAAI 2026

First

Dropouts in Confidence: Moral Uncertainty in Human-LLM Alignment

Kwon et al.

ACL 2026 (*Findings*)

Co-corresponding

Machine Behavior in Relational Moral Dilemmas: Moral Rightness, Predicted Human Behavior, and Model Decisions

Kim, Kwon et al.

NeuroAI

NeurIPS 2023

First

Transformer as a Hippocampal Memory Consolidation Model Based on NMDAR-Inspired Nonlinearity

Kim, Kwon et al.

ICLR 2025

First

Brain-inspired Lp-Convolution Benefits Large Kernels and Aligns Better with Visual Cortex

Kwon et al.

IJCV 2024

First

SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior

Kwon et al.

Exp. Mol. Med. 2026

First

Cerebellar Tonic Inhibition Orchestrates the Maturation of Information Processing and Motor Coordination

Kwon et al.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher in Trustworthy AI	2024 - Present
Max Planck Institute for Security and Privacy (MPI-SP), Bochum, Germany	
◊ Advisor: Meeyoung Cha, Scientific Director of MPI-SP	
Postdoctoral Researcher in NeuroAI	2022 - 2024
Institute for Basic Science (IBS), Daejeon, South Korea	
Researcher in Neuroscience	2018 - 2020
Institute for Basic Science (IBS), Daejeon, South Korea	
Student Researcher	2014 - 2018
Korea Institute of Science and Technology (KIST), Seoul, South Korea	

EDUCATION

Ph.D. in Nano-Bio-Information-Technology	2014 - 2022
Korea University, Seoul, South Korea	
◊ Advisor: C. Justin Lee, Director of IBS	
◊ Full-Ride Scholarship, KU-KIST Graduate School	
B.E. in Electrical and Electronic System	2008 - 2012
Saitama University, Saitama, Japan	
◊ Full-Ride Scholarship, Korea-Japan Governments Joint Scholarship Program	

PUBLICATION SUMMARY

In Conferences

- *main author* (10): ICML2026, AAAI2026, ICLR2025, NeurIPS2023, ICML2026-w, ACL2026-f, ICLR2025-w, ICLR2024-w, ICLR2024-w, CVPR2023-w
- *co-author* (8): ICML2026, ACL2026, ICLR2026, ICLR2026, ICLR2026-w, ICLR2025-b, NeurIPS2022-w, CVPR2022-w

In Journals

- *main author* (6): Exp.Mol.Med.2026, IJCV2024, Exp.Neurobiol.2024, Mol.Brain2021, Exp.Neurobiol.2018, Exp.Neurobiol.2017
- *co-author* (14): Nat.Commun.2025, Nat.Commun.2024, Nat.Metab.2023, Brain2023, Mol.Brain2022, CellMetab.2022, Exp.Neurobiol.2022, Exp.Neurobiol.2021, Exp.Neurobiol.2021, J.Neurosci.2020, Curr.Biol.2019, Exp.Neurobiol.2019, PNAS2018, Nat.Commun.2016

Citations: 1028 · h-index 12 · i10-index 17

as of June 2026

Suffixes: -w = workshop, -f = findings, -b = blogpost

ACHIEVEMENTS

Seal of Excellence — MSCA Postdoctoral Fellowships 2025 · European Commission, Horizon Europe (HORIZON-MSCA-2025-PF-01-01) · 2026 *Proposal 101286042 — TRACE: Towards Reliable AI through Cognitive Engrams (with Max-Planck-Gesellschaft)*, total score 89.40%

Excellent Paper Award · Korean Artificial Intelligence Association (JKAIA) · 2023 *Brain-inspired Lp-Convolution benefits large kernels and aligns better with visual cortex*

Patent Application & Technology Transfer • Application Number: 10-2023-0027219 • 2023
Device and Method for Classification of Human and Animal Behavior, Patent Contribution=35%;
Finalized a technology transfer with ACTNOVA Inc.

Best Presentation Award • Korean Society for Brain and Neural Sciences (KSBNS) • 2022
ABCD-analysis: Mapping animal behavior and differential analysis of kinematic features without selection bias

Young Investigator Award • Korean Society for Molecular and Cellular Biology (KSMCB)
• 2022 *Retina-attached slice recording reveals light-triggered tonic GABA signaling in suprachiasmatic nucleus*

Best Poster Award • Center for Cognition and Sociality (CCS) Workshop • 2021 *Mapping Behavior with DECLARE: Deep Embedded Clustering of Action REpresentation*

TALKS

AI Engram: In Search of Memory Traces in Artificial Neural Networks • Scalable Trustworthy AI (STAI) group - KAIST • 2026 (Invited Seminar, In person, invited by Seong Joon Oh)

AI Engram: In Search of Memory Traces in Artificial Neural Networks • Max Planck Institute for Software Systems • 2026 (Invited Seminar, In person, invited by Krishna Gummadi)

AI Engram: In Search of Memory Traces in Artificial Neural Networks • Spark(l)ing Science - Max Planck Institute for Security and Privacy • 2026 (Seminar, In person)

How neuroscience can benefit AI • KAIST BK21 IBS Symposium - Integrated Neuroscience and Physiology • 2025 (Invited Seminar, In person, invited by Greg Suh)

Transformer as a Hippocampal Memory Consolidation Model based on NMDAR-inspired Nonlinearity • KSBNS Symposium • 2025 (Invited Seminar, In person)

Sovereign AI: K-AI as an opportunity • Netzwerk Junge Generation Deutschland-Korea • 2025 (Public Lecture, In person, invited by Muhong Lee)

Brain-inspired AI • Institute for Basic Science (IBS) • 2025 (Invited Seminar, In person, invited by C. Justin Lee)

Brain-inspired AI • Korea Brain Research Institute (KBRI) • 2025 (Invited Seminar, In person, invited by Jeongyeon Kim)

Brain-inspired AI • Gwangju Institute of Science and Technology (GIST) • 2025 (Invited Seminar, In person, invited by Sundong Kim)

Neuroscience for Artificial Intelligence • MLAI group, Yonsei University • 2025 (Invited Lecture, Offline, invited by Kyungwoo Song)

Brain-inspired AI • Kyungpook National University (KNU) • 2024 (Invited Seminar, In person, invited by Saekwang Nam)

SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior • Korea Institute of Ocean Science & Technology (KIOST) • 2024 (Invited Seminar, In person, invited by Soomee Kim)

SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior • Korea Brain Research Institute (KBRI) • 2024 (Invited Seminar, In person, invited by Jeongyeon Kim)

Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity • IBS-CCS Symposium • 2023 (Oral Presentation, In person)

Memory editing in the brain - Eternal Sunshine of the Spotless Mind • IBS Science in the Cinema Roof • 2023 (Public Lecture, In person)

From human mind to artificial intelligence - Is ChatGPT a bridge • Reading Festival at Jeonil High School • 2023 (Public Lecture, In person)

Resolving clustering bias during animal behavior mapping and kinematic profiling • IBS Data Science Group seminar • 2022 (Invited Seminar, Online, invited by Meeyoung Cha)

Resemblances Between Transformer's Nonlinearity and NMDA Receptor Dynamics • IBS Winter School on AI-Boosted Basic Science • 2022 (Invited Lecture, In person)

Age-dependent motor coordination analysis by 3D pose-estimation using AVATAR • IBS 2021 2nd Workshop • 2021 (Oral Presentation, In person)

Backpropagation and the Brain • KAIST-NeuroAI Journal Club • 2020 (Oral Presentation, Hybrid)

Deep learning with NMDA receptor inspired activation function • IBS 2019 Year-End Workshop • 2020 (Oral Presentation, In person)

Development of a Low-cost, Comprehensive Recording System for Circadian Rhythm Behavior • Korean Society for Molecular and Cellular Biology (KSMCB) • 2019 (Oral Presentation, In person)

TEACHING

Guest Lecturer • Georgia Tech CSE 8803 BMI "Brain-inspired Machine Intelligence", Fall 2024

Guest Lecturer • Ewha Womans University G 18067 "Introduction to System Health", Spring 2026

Mentoring • SoHyung Kim (Korea University, Undergraduate Intern), IBS 2020, 6 months • Outputs: CVPR2023-w, IJCV2024 (co-author)

Mentoring • Hyewon Kim (Eulji University School of Medicine, Undergraduate Intern), IBS 2024, 9 months • Outputs: ICLR2024-w (co-author), Exp.Neurobiol.2024 (co-first author)

Mentoring • Minsung Kim (Seoul National University, Ph.D. Intern, with Dong-Kyum Kim), MPI-SP 2024, 3 months • Outputs: ICLR2025-b, ACL2026 (first author)

Mentoring • Jiseon Kim (KAIST, Ph.D. Intern, with Luiz Felipe Vecchiatti), MPI-SP 2025, 6 months • Outputs: ICLR2025-w, ACL2026-f (first author)

Mentoring • Nakyeong Yang (Seoul National University, Ph.D. Intern, with Dong-Kyum Kim), MPI-SP 2025, 3 months • Outputs: ICLR2026 (first author)

COMMUNITY SERVICE

Symposium Organizer • KSBNS 2026 Annual Meeting • "Minds Meet Machines: Bidirectional Innovations in AI and Neuroscience"

Symposium Co-Organizer • KSBNS & CJK 2025 Joint Symposium • "Neuroscience-inspired AI: Computational Insights into Biological and Artificial Intelligence"

Program Committee • AAAI 2026, AI Alignment Track

Program Committee • Machine+Behavior 2026, Harnack House, Berlin

Emergency Reviewer • ICML 2026

Reviewer • NeurIPS 2026

Reviewer • ICML 2026 Mechanistic Interpretability Workshop

Reviewer • ICLR 2026

Reviewer • ICLR 2026 Workshop on Representational Alignment (Re-Align)

Reviewer • ICLR 2025 Workshop on Representational Alignment (Re-Align)

Founding Member • KAIST-NeuroAI Journal Club, 2020

PUBLICATIONS

✓ selected publication * first author # corresponding author

- ✓ **J. Kwon***, J. Kim, D. Kim, M. Cha. Moir: Let the Model Direct Its Own Story for Robust Cross-Domain Knowledge Editing. *ICML Mechanistic Interpretability Workshop*, 2026.
- ✓ **J. Kwon***, D. Kim*, J. Kim*, Y. Kim, W. Kook, M. Cha. AI Engram: In Search of Memory Traces in Artificial Intelligence. *International Conference on Machine Learning*, 2026 (Oral).
M. S. Locatelli, F. Tonucci, **J. Kwon**, L. F. Vecchietti, B. N. Wijaya, C. Y. Low, V. Almeida, M. Cha. Textual Supervision Enhances Geospatial Representations in Vision-Language Models. *International Conference on Machine Learning*, 2026 (Acceptance Rate=26.6%).
- ✓ J. Kim, **J. Kwon#**, L. F. Vecchietti#, W. Dong, J. Kim, M. Cha#. Machine Behavior in Relational Moral Dilemmas: Moral Rightness, Predicted Human Behavior, and Model Decisions. *Findings of the 64th Annual Meeting of the Association for Computational Linguistics*, 2026 (Acceptance Rate=18%).
M. Kim, D. Kim, **J. Kwon**, N. Yang, K. Jung, M. Cha. How Training Data Shapes the Use of Parametric and In-Context Knowledge in Language Models. *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, 2026 (Acceptance Rate=19%).
S. Kim, **J. Kwon**. From Human Cognition to AI Reasoning: Models, Methods, and Applications. *ICLR HCAIR Workshop*, 2026.
- ✓ **J. Kwon***, S. Kim*, J. Woo, K. Tanaka-Yamamoto, O. James, E. De Schutter, S. Hong, C. J. Lee. Cerebellar Tonic Inhibition Orchestrates the Maturation of Information Processing and Motor Coordination. *Experimental & Molecular Medicine*, 2026 (5-year IF=14.2, 2024).
- ✓ **J. Kwon***, L. F. Vecchietti, S. Park, M. Cha. Dropouts in Confidence: Moral Uncertainty in Human-LLM Alignment. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026 (AI Alignment Track, Acceptance Rate=22.65%).
D. Kim, M. Kim, **J. Kwon**, N. Yang, M. Cha. Bilinear Relational Structure Fixes Reversal Curse and Enables Consistent Model Editing. *International Conference on Learning Representations*, 2026 (Acceptance Rate=28%).
N. Yang, D. Kim, **J. Kwon**, M. Kim, K. Jung, M. Cha. Erase or Hide? Suppressing Spurious Unlearning Neurons for Robust Unlearning. *International Conference on Learning Representations*, 2026 (Acceptance Rate=28%).
M. Kim, **J. Kwon**, D. Kim, M. Cha. In Search of the Engram in LLMs: A Neuroscience Perspective on the Memory Functions in AI Models. *ICLR Blogposts Track*, 2025.
B. Hyeon, J. Shin, J. Lee, W. Kim, **J. Kwon**, H. Lee, D. Kim, C. Y. Kim, S. Choi, J. Jeong, K. Kim, C. J. Lee, D. Kim, W. D. Heo. Integrating Artificial Intelligence and Optogenetics for Parkinson's Disease Diagnosis and Therapeutics in Male Mice. *Nature Communications*, 2025.
J. Kim*, **J. Kwon***, L. F. Vecchietti*, A. Oh, M. Cha. Exploring Persona-dependent LLM Alignment for the Moral Machine Experiment. *ICLR Bi-Align Workshop*, 2025.
- ✓ **J. Kwon***, S. Lim, K. Song, C. J. Lee. Brain-inspired Lp-Convolution Benefits Large Kernels and Aligns Better with Visual Cortex. *International Conference on Learning Representations*, 2025 (Acceptance Rate=32.08%).
J. Kwon*, M. Sa, H. Kim, Y. Seong, C. J. Lee. Egocentric 3D Skeleton Learning in Identity-Aware Deep LSTM Network Encodes Obese-Like Motion Representations. *ICLR TS4H Workshop*, 2024.
J. Kwon*, S. Lim, K. Song, C. J. Lee. Brain-inspired Lp-Convolution Benefits Large Kernels and Aligns Better with Visual Cortex. *ICLR Re-Align Workshop*, 2024.
- ✓ **J. Kwon***, S. Kim, D. Kim, J. Joo, S. Kim, M. Cha, C. J. Lee. SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior. *International Journal of Computer Vision*, 2024 (5-year IF=15.5, 2024).
- ✓ **J. Kwon***, M. Sa, H. Kim, Y. Seong, C. J. Lee. Egocentric 3D Skeleton Learning in a Deep Neural Network Encodes Obese-like Motion Representations. *Experimental Neurobiology*, 2024

(Featured on the cover).

H. Kang, A. Han, A. Zhang, H. Jeong, W. Koh, J. M. Lee, H. Lee, H. Y. Jo, M. A. Maria-Solano, M. Bhalla, **J. Kwon**, et al.. GolpHCat (TMEM87A), a Unique Voltage-Dependent Cation Channel in Golgi Apparatus, Contributes to Golgi-pH Maintenance and Hippocampus-Dependent Memory. *Nature Communications*, 2024.

- ✓ D. Kim*, **J. Kwon***, M. Cha, C. J. Lee. Transformer as a Hippocampal Memory Consolidation Model Based on NMDAR-Inspired Nonlinearity. *Advances in Neural Information Processing Systems*, 2023 (Acceptance Rate=26.1%).

M. Sa, E. Yoo, W. Koh, M. G. Park, H. Jang, Y. R. Yang, M. Bhalla, J. Lee, J. Lim, W. Won, **J. Kwon**, et al.. Hypothalamic GABRA5-Positive Neurons Control Obesity via Astrocytic GABA. *Nature Metabolism*, 2023.

J. Kwon*, S. Kim, D. Kim, J. Joo, S. Kim, M. Cha, C. J. Lee. SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior. *CVPR CV4Animals Workshop*, 2023.

M. Nam, H. Y. Ko, D. Kim, S. Lee, Y. M. Park, S. J. Hyeon, W. Won, J. Chung, S. Y. Kim, H. H. Jo, K. T. Oh, Y. Han, G. Lee, Y. H. Ju, H. Lee, H. Kim, J. Heo, M. Bhalla, K. J. Kim, **J. Kwon**, et al.. Visualizing Reactive Astrocyte-Neuron Interaction in Alzheimer's Disease Using ¹¹C-Acetate and ¹⁸F-FDG. *Brain*, 2023.

D. Kim, **J. Kwon**, M. Cha, C. J. Lee. Transformer Needs NMDA Receptor Nonlinearity for Long-Term Memory. *NeurIPS MemARI Workshop*, 2022.

D. Kim, J. Kim, W. Jung, J. Park, M. Kim, A. Shin, Y. Jeong, S. Park, G. Shin, Y. W. Lee, **J. Kwon**, D. Kim. AVATAR: AI Vision Analysis for Three-Dimensional Action in Real-Time. *CVPR CV4Animals Workshop*, 2022.

S. Kim, **J. Kwon**, M. G. Park, C. J. Lee. Dopamine-Induced Astrocytic Ca²⁺ Signaling in mPFC is Mediated by MAO-B in Young Mice, but by Dopamine Receptors in Adult Mice. *Molecular Brain*, 2022.

Y. H. Ju, M. Bhalla, S. J. Hyeon, J. E. Oh, S. Yoo, U. Chae, **J. Kwon**, W. Koh, J. Lim, Y. M. Park, et al.. Astrocytic Urea Cycle Detoxifies A β -Derived Ammonia While Impairing Memory in Alzheimer's Disease. *Cell Metabolism*, 2022.

J. M. Lee, M. Sa, H. An, J. M. J. Kim, **J. Kwon**, B. Yoon, C. J. Lee. Generation of Astrocyte-Specific MAOB Conditional Knockout Mouse with Minimal Tonic GABA Inhibition. *Experimental Neurobiology*, 2022.

- ✓ **J. Kwon***, M. W. Jang, C. J. Lee. Retina-Attached Slice Recording Reveals Light-Triggered Tonic GABA Signaling in Suprachiasmatic Nucleus. *Molecular Brain*, 2021.

M. W. Jang, T. Y. Kim, K. Sharma, **J. Kwon**, E. Yi, C. J. Lee. A Deafness Associated Protein TMEM43 Interacts with KCNK3 (TASK-1) Two-Pore Domain K⁺ (K2P) Channel in the Cochlea. *Experimental Neurobiology*, 2021.

J. Won, H. H. Kazan, **J. Kwon**, M. Park, M. A. Ergun, S. Ozcan, B. Y. Choi, W. D. Heo, C. J. Lee. Ultimate COVID-19 Detection Protocol Based on Saliva Sampling and qRT-PCR with Risk Probability Assessment. *Experimental Neurobiology*, 2021.

K. Han, M. Lee, H. Lim, M. W. Jang, **J. Kwon**, C. J. Lee, S. Kim, M. Suh. Excitation-Inhibition Imbalance Leads to Alteration of Neuronal Coherence and Neurovascular Coupling Under Acute Stress. *Journal of Neuroscience*, 2020.

S. Oh, J. M. Lee, H. Kim, J. Lee, S. Han, J. Y. Bae, G. Hong, W. Koh, **J. Kwon**, E. Hwang, et al.. Ultrasonic Neuromodulation via Astrocytic TRPA1. *Current Biology*, 2019.

Y. Han, **J. Kwon**, J. Won, H. An, M. W. Jang, J. Woo, J. S. Lee, M. G. Park, B. Yoon, S. E. Lee, et al.. Tweety-Homolog (Ttyh) Family Encodes the Pore-Forming Subunits of the Swelling-Dependent Volume-Regulated Anion Channel (VRAC_{swell}) in the Brain. *Experimental Neurobiology*, 2019.

J. Woo, J. O. Min, D. Kang, Y. S. Kim, G. H. Jung, H. J. Park, S. Kim, H. An, **J. Kwon**, J. Kim, et al.. Control of Motor Coordination by Astrocytic Tonic GABA Release Through Modulation of Excitation/Inhibition Balance in Cerebellum. *Proceedings of the National Academy of Sciences*, 2018.

J. Kwon*, M. G. Park, S. E. Lee, C. J. Lee. Development of a Low-Cost, Comprehensive Recording System for Circadian Rhythm Behavior. *Experimental Neurobiology*, 2018.

J. Kwon*, H. An, M. Sa, J. Won, J. I. Shin, C. J. Lee. Orai1 and Orai3 in Combination with Stim1 Mediate the Majority of Store-Operated Calcium Entry in Astrocytes. *Experimental Neurobiology*, 2017.

G. E. Ha, J. Lee, H. Kwak, K. Song, **J. Kwon**, S. Jung, J. Hong, G. Chang, E. M. Hwang, H. Shin, et al.. The Ca²⁺-Activated Chloride Channel Anoctamin-2 Mediates Spike-Frequency Adaptation and Regulates Sensory Transmission in Thalamocortical Neurons. *Nature Communications*, 2016.